

EECE 5550 Mobile Robotics Final Project

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Due: Thursday December 11, 2025 by midnight

Project Description

One of the most important skills for working in a fast-moving field (such as robotics) is the ability to search through the literature effectively to identify techniques that are relevant for a problem that you would like to solve, implement those methods yourself, and (crucially) identify areas where the current state-of-the-art may still not be entirely satisfactory. In this project, you will have an opportunity to practice these skills while engaging more deeply with some of the material that we have been discussing in the class. You will have two options:

1. Choose a recent algorithmic paper from a top robotics research venue (e.g. T-RO, IJRR, RA-L, ICRA, RSS, IROS, etc.) that you and your team are interested in learning more about, and conduct a **rigorous** empirical evaluation of its performance: does the method *really* perform the way that the paper claims it does? (What happens if you run it on a new set of test problems? Did the authors of the original paper “overfit” their method to their own data sets? Are there obvious problem parameters, such as the level of noise on sensor data, whose effects the authors did not sufficiently characterize?) Based upon these results, you should suggest (and try out) some potential extensions / improvements of the original algorithm, and evaluate their efficacy.
2. Formulate an interesting *nontrivial* problem of your own choosing — perhaps one from your own research — that is relevant to some aspect of the material that we have discussed in the course, and that is not adequately addressed in the existing literature.¹ Explain why existing methods do not satisfactorily address your problem, and propose a new algorithm of your own design for solving it. Conduct a rigorous empirical evaluation of your proposed method’s performance on your target task, and assess its efficacy.

Project Scope

The paper / problem that you choose should focus on a topic that is relevant to this class; your instructor will have to approve the choice to make sure that it is a good fit. Some potential topics can be (but are not limited to):

1. SLAM (initialization, optimization, outlier-robust methods, approximate inference methods, certifiable methods ...)
2. Distributed optimization for multi-agent systems
3. Place recognition / loop closure detection

¹Note that establishing that there is indeed a gap in the existing literature requires a *thorough* lit review.

4. Shape estimation (using e.g. Gaussian process regression, neural implicit models, etc.)
5. Optical flow-based camera tracking / structure estimation
6. Visual-inertial state estimation
7. Object detection
8. Dynamic programming to solve decision-making problems with MDPs
9. Model predictive control for scalable autonomous decision-making/control synthesis
10. Control barrier functions to ensure constraint satisfaction
11. Sampling based methods for trajectory planning

Proposal

Your team is expected to provide a few sentences or short paragraphs (**by Sunday November 9th**) answering the following questions:

- What are you proposing to do?
 - Option 1: Specify the which paper / algorithm you propose to study, and what aspects of its performance you propose to investigate. Why did you pick *this specific* paper? [NB: Usually there are *many* papers on a given topic (e.g. optical flow), of highly-varying quality. Don't simply believe the first one you happen to come across!]
 - Option 2: Write down a description of the problem that you propose to study, and what related papers / methods you will include in your literature review.

In both cases, you should include **precise mathematical descriptions** of:

- The problem that your algorithm(s) are addressing.
- What *specific* experiment(s) you plan to run, including:
 - * What *specific* phenomena you are interested in studying / measuring
 - * How you plan to acquire or generate any data that you need to run these experiments
 - * How you will *quantify* these experimentally

Your description should be sufficiently detailed that I could *exactly* reproduce your experiments.

- What do you expect the output of your project to be (i.e. what would a successful execution of your project look like)? Please be as specific as possible.

For example: “I will produce a piece of code that will accept as input X and return Y, thereby solving Problem P.”

Final Deliverables

Your team will do a **group presentation** in the last week of lectures to present the proposed topic and show preliminary results. This will be an opportunity to get feedback about your work before the final report submission.

Moreover, your team should submit the following by **Thursday December 12th**:

1. The code that your team implements or modifies.
2. A group report describing the project. This does not need to be lengthy, but it must contain the following elements:
 - Use IEEE conference template for your report (<https://www.overleaf.com/latex/templates/ieee-conference-template/grfzhhncsfqn>)
 - Abstract: A brief executive summary of your project report
 - Introduction and Motivation: explain why you choose that paper / problem
 - Problem Statement: A description of the problem that you are studying, including a **mathematically precise formulation**
 - Proposed Solution: A **mathematically precise** description of the method that you implemented. [NB: This should be sufficiently detailed and specific that I could *exactly reproduce* your code.]
 - Results: Conduct a **rigorous, quantitative assessment** of your method's performance, and report its quantitative performance measures.
 - Conclusion: Summarize your major findings, including your algorithm's limitations, potential extensions, major technical challenges faced, etc.
 - References